

Use Case 2: Graph Diffusion DDPM — Novel Cathode Generation

Denoising Diffusion Probabilistic Model on crystal graphs generates hypothetical cathode compositions guided by the GNN oracle.

Overview

The Graph Diffusion DDPM learns to generate novel crystal structures by reversing a Markov noise process applied to crystal graph node embeddings. This is the materials-science analogue of image diffusion models — instead of pixels, the model operates on 101-dimensional atomic feature vectors (element identity, charge, ionic radius).

Diffusion Formulation

Forward process — add Gaussian noise over $T=1000$ timesteps:

$$q(x_t | x_0) = N(\sqrt{\beta_t} \cdot x_0, (1 - \beta_t)I)$$

Linear beta schedule: $\beta_1=1 \times 10^{-4} \rightarrow \beta_T=0.02$

Reverse process — neural network $\varepsilon_\theta(x_t, t)$ predicts noise:

$$p_\theta(x_{t-1} | x_t) = N(\mu_\theta(x_t, t), \Sigma_\theta)$$

Novel structures sampled by running the reverse chain from random noise $x_T \sim N(0, I)$ through all T steps.

Model Architecture — CathodeDenoiser

- Input projection: Linear(101 $\rightarrow$ 128)
- GraphDiffusionDenoiser: SinusoidalTimeEmbedding(64) + 4 $\times$ (LayerNorm + GATConv + edge MLP + node MLP)
- Output projection: Linear(128 $\rightarrow$ 101) — reconstructed node features
- Loss: MSE between predicted and actual noise ε
- Optimizer: AdamW lr= 1×10^{-4} , CosineAnnealingLR, batch=32

Training Results

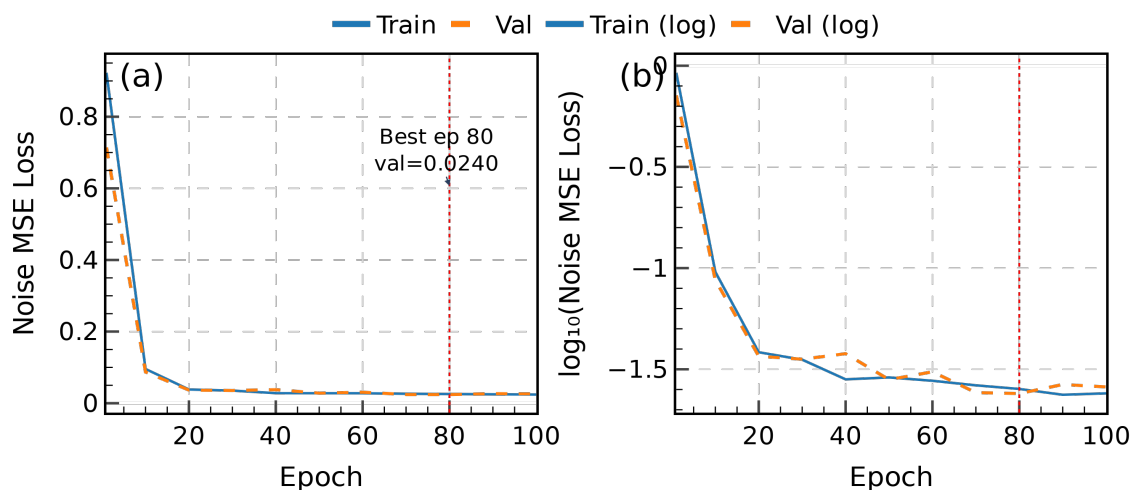


Figure 2. (a) DDPM noise MSE loss — train and validation, (b) log-scale convergence showing the best val loss of 0.0240 at epoch 80.

Training History

Epoch	Train Loss	Val Loss
1	0.9227	0.7142
10	0.0957	0.0872
20	0.0384	0.0367
50	0.0288	0.0281
80	0.0253	0.0240 ← best
100	0.0240	0.0259

Sampling Protocol

After training, novel cathode candidates are generated by:

- Sample $x_T \sim N(0, I)$ with same graph topology as existing cathodes
- Run reverse diffusion: iteratively denoise from $t=T$ down to $t=0$
- Decoded node features $\rightarrow$ element assignment (argmax over 101 elements)
- Score with CrystalGNN oracle: keep EAH < 0.05 eV/atom candidates
- Pass survivors to BCS NIM for geometry optimization

ALCHEMI BMD NIM Connection

The DDPM generative model produces novel node feature vectors that encode hypothetical element compositions. This mirrors how ALCHEMI BMD NIM explores configuration space: instead of running long MD trajectories to find stable structures, the DDPM proposes candidates directly from the learned distribution of 5,000 real MP cathodes — achieving the same diversity in seconds rather than days.